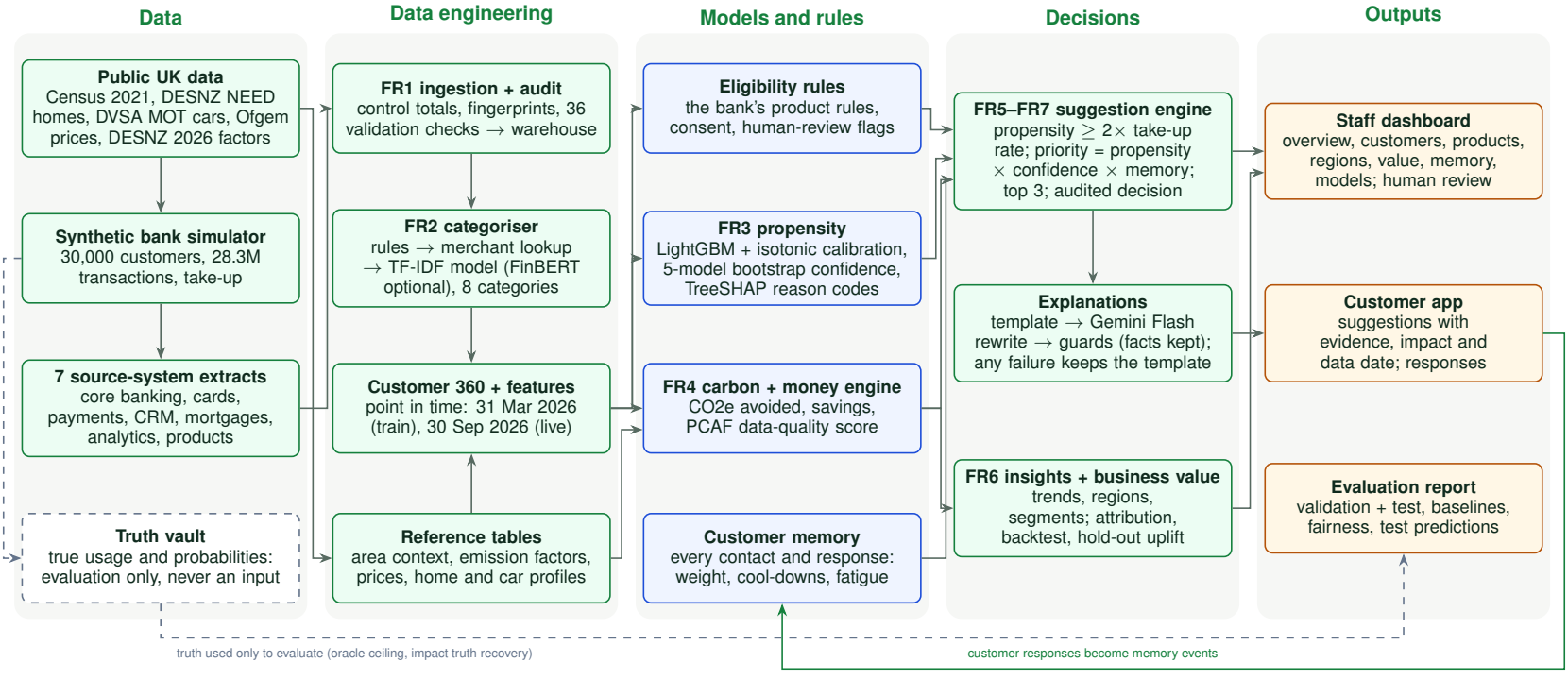
Data
Public UK data
Census 2021, DESNZ NEED homes, DVSA MOT cars, Ofgem prices, DESNZ 2026 factors
Synthetic bank simulator
30,000 customers, 28.3M transactions, take-up
7 source-system extracts
core banking, cards, payments, CRM, mortgages, analytics, products
Truth vault
true usage and probabilities: evaluation only, never an input
Data engineering
FR1 ingestion + audit
control totals, fingerprints, 36 validation checks → warehouse
FR2 categoriser
rules → merchant lookup → TF-IDF model (FinBERT optional), 8 categories
Customer 360 + features
point in time: 31 Mar 2026 (train), 30 Sep 2026 (live)
Reference tables
area context, emission factors, prices, home and car profiles
Models and rules
Eligibility rules
the bank's product rules, consent, human-review flags
FR3 propensity
LightGBM + isotonic calibration, 5-model bootstrap confidence, TreeSHAP reason codes
FR4 carbon + money engine
CO2e avoided, savings, PCAF data-quality score
Customer memory
every contact and response: weight, cool-downs, fatigue
Decisions
FR5–FR7 suggestion engine
propensity ≥ 2× take-up rate; priority = propensity × confidence × memory; top 3; audited decision
Explanations
template → Gemini Flash rewrite → guards (facts kept); any failure keeps the template
FR6 insights + business value
trends, regions, segments; attribution, backtest, hold-out uplift
Outputs
Staff dashboard
overview, customers, products, regions, value, memory, models; human review
Customer app
suggestions with evidence, impact and data date; responses
Evaluation report
validation + test, baselines, fairness, test predictions
truth used only to evaluate (oracle ceiling, impact truth recovery)
customer responses become memory events